

[Home](#) > [Prospective Students](#) > [Majors](#) > BS CSE Program Information

BS CSE Program Information

The Bachelor of Science in Computer Science and Engineering offers students a general education in engineering, physical sciences, and mathematics, along with intensive study in computer science.

> Computer Science and Non-Major Course Requirements

CSE Core



All BS CSE students must take the set of eight courses outlined below:

Core CSE Courses

Course	Title	Credits	Prerequisites and Notes
CSE 2221	Software I: Software Components	4	Must have credit for CSE 1222, CSE 1223, CSE 1224, or ENGR 1281, or have passed the 2221 placement test
CSE 2231	Software II: Software Development and Design	4	Must have credit for CSE 2221 and Math 1151
CSE 2321	Foundations I: Discrete Structures	3	Must have credit for CSE 2221 and Math 1151

Course	Title	Credits	Prerequisites and Notes
CSE 2331	Foundations II: Data Structures and Algorithms	3	Must have credit for CSE 2231 and CSE 2321, and STAT 3470
CSE 2421	Systems I: Introduction to Low-Level Programming	4	Must have credit for CSE 2231 and CSE 2321
CSE 2431	Systems II: Introduction to Operating Systems	3	Must have credit for CSE 2421
CSE 3341	Principles of Programming Languages	3	Must have credit for CSE 2331, CSE 2421 and CSE 3901, CSE 3902 or CSE 3903
CSE 2501 or Philos 2338*	Social, Ethical, and Professional Issues in Computing or Ethics in the Professions: Introduction to Computing Ethics and Effective Presentation	1	For CSE 2501, must have credit for CSE 2231, CSE 2321, and GE foundation writing and info literacy course

*For students who entered the program in or after AU22. Students who entered prior to AU22 or have permission to take the Legacy General Education would take Philos 1338.

CSE Core Choices



In addition to the core CSE courses, which are designed to provide a breadth of knowledge, BS CSE students choose five additional courses. Students have the option to choose these courses independently or to use these courses to fulfill some of their specialization requirements. The decisions BS CSE students must make are as follows:

Software Engineering or Databases

Course	Title	Credits	Prerequisites and Notes	Specialization Requirement
CSE 3231	Software Engineering Techniques	3	Must have credit for CSE 3901, CSE 3902, or CSE 3903. Not open to students with credit for 5231.	Software Engineering

Course	Title	Credits	Prerequisites and Notes	Specialization Requirement
CSE 3241	Introduction to Database Systems	3	Must have credit for CSE 2123, CSE 2231, CSE 2321, or CSE 2111 and Math 2366. Not open to students with credit for 5241.	Database Systems and Data Analytics

Computer Architecture or Networking

Course	Title	Credits	Prerequisites and Notes	Specialization Requirement
CSE 3421	Introduction to Computer Architecture	3	Must have credit for CSE 2231, and CSE 2421 or ECE 2560, and ECE 2060. Not open to students with credit for 5421.	Computer Systems
CSE 3461	Computer Networking and Internet Technologies	3	Must have credit for CSE 2421; or CSE 3430; or CSE 2451 and ECE 2560. Not open to students with credit for 5461.	Information and Computation Assurance, Computer Networking, Computer Systems (elective)

AI or Game and Animation

Course	Title	Credits	Prerequisites and Notes	Specialization Requirement
CSE 3521	Survey of Artificial Intelligence I: Basic Techniques	3	Must have credit for CSE 2331 or CSE 5331, Math 2174 or 2568 or 4568 or 5520H, and Stat 3201 or 3450 or 3460 or 3470 or 4201 or Math 4530 or 5530H. Not open to students with credit for 5521.	Artificial Intelligence
CSE 3541	Computer Game and Animation Technique	3	Must have credit for CSE 3901, 3902, or 3903. Not open to students with credit for 5541.	Computer Graphics and Game Design

Project

Course	Title	Credits	Prerequisites and Notes	Specialization Requirement
CSE 3901	Project: Design, Development, and Documentation of Web Applications	4	Must have credit for CSE 2231; and 2321; and 2421, or 3430, or 2451 and ECE 2560.	
CSE 3902	Project: Design, Development, and Documentation of Interactive Systems	4	Must have credit for CSE 2231; and 2321; and 2421, or 3430, or 2451 and ECE 2560	Computer Graphics and Game Design
CSE 3903	Project: Design, Development, and Documentation of System Software	4	Must have credit for CSE 2231; and 2321; and 2421, or 3430, or 2451 and ECE 2560	

Capstone

Course	Title	Credits	Prerequisites and Notes	Specialization Requirement
CSE 5911	Capstone Design: Software Applications	4	Must have credit for CSE 3231, or 5231; and 2501, or Philos 1338, or 2338; and CSE 3901, 3902, or 3903; and GE writing and comm course: level 2, or GE foundation writing and info literacy course.	
CSE 5912	Capstone Design: Game Design and Development	4	Must have credit for CSE 3541, or 5541; and 2501, or Philos 1338, or 2338; and CSE 3901, or 3902, or 3903; and GE writing and comm course: level 2, or GE foundation writing and info literacy course.	Computer Graphics and Game Design (elective)
CSE 5913	Capstone Design: Computer Animation	4	Must have credit for CSE 3541, or 5541; and 2501, or Philos 1338, or 2338; and CSE 3901, or 3902, or 3903; and GE writing and comm course: level 2; or GE foundation writing and info literacy course.	Computer Graphics and Game Design (elective)

Course	Title	Credits	Prerequisites and Notes	Specialization Requirement
CSE 5914	Capstone Design: Knowledge-Based Systems	4	Must have credit for CSE 3521, or 5521; and 2501, or Philos 1338, or 2338; and CSE 3901, or 3902, or 3903; and GE writing and comm course: level 2, or GE foundation writing and info literacy course.	Artificial Intelligence (elective)
CSE 5915	Capstone Design: Information Systems	4	Must have credit for CSE 3241, or 5241; and 2501, or Philos 1338, or 2338; and CSE 3901, or 3902, or 3903; and GE writing and comm course: level 2, or GE foundation writing and info literacy course.	

Technical Electives and Specializations



Technical electives

All BS CSE students must choose a **specialization option**, which will determine a portion of a student's technical electives. Overall, each student must have 17 credit hours of technical elective coursework, of which 9 hours must be in CSE. Some of these 9 hours will be determined by the selected specialization. The remaining 8 hours of technical electives may be additional CSE course at the 3000-level or higher not already counting toward a degree requirement or select non-CSE courses at the 2000-level or higher.

Limitations for certain CSE courses

Students are limited in the number of hours they may take of CSE courses that are graded S/U or that focus on research.

Course	Title	Credit	Prerequisites and Notes
CSE 4251	The UNIX Programming Environment	1	Must have credit for CSE 2122, 2123, or 2231; and 2321; and enrollment in CSE, CIS, ECE, Engr Physics, or Data Analytics major, or CS minor. This course is graded S/U. No more

Course	Title	Credit	Prerequisites and Notes
			than 2 of CSE 4251, 4252, 4253, and 4256 may be counted toward technical electives.
CSE 4252	Programming in C++	1	Must have credit for CSE 2123 or 2231; and 2321; and enrollment in CSE, CIS, ECE, Engr Physics, or Data Analytics major, or CS minor. Not open to students with credit for 2122. This course is graded S/U. No more than 2 of CSE 4251, 4252, 4253, and 4256 may be counted toward technical electives. No more than 2 of CSE 4251, 4252, 4253, and 4256 may be counted toward technical electives.
CSE 4253	Programming in C#	1	Must have credit for CSE 2122, 2123, or 2231; and 2321; and enrollment in CSE, CIS, ECE, Engr Physics, or Data Analytics major, or CS minor. This course is graded S/U. No more than 2 of CSE 4251, 4252, 4253, and 4256 may be counted toward technical electives.
CSE 4256	Programming in Python	1	Must have credit for CSE 2122, 2123, or 2231; and 2321; and enrollment in CSE, CIS, ECE, Engr Physics, or Data Analytics major, or CS minor. This course is graded S/U. No more than 2 of CSE 4251, 4252, 4253, and 4256 may be counted toward technical electives.
CSE 4193	Individual Studies in Computer Science and Engineering	up to 2	Must have permission of instructor. This course is graded S/U. No more than 6 hours total of CSE 4193, 4998, and 4999 combined may be counted toward technical electives.
CSE 4998	Undergraduate Research in Computer Science and Engineering	up to 3	Permission of instructor. This course is graded S/U. No more than 6 hours total of CSE 4193, 4998, and 4999 combined may be counted toward technical electives.
CSE 4999	Computer Science and Engineering Research for Thesis	up to 6	Permission of instructor. This course is graded S/U. No more than 6 hours total of CSE 4193, 4998, and 4999 combined may be counted toward technical electives.

Non-CSE technical electives

If you are looking to use non-CSE course as part of your technical electives, you are allowed to do so if you met one of the following qualifications:

- The course is a part of any completed minor.
- The course is listed on [the pre-approved list](#).
- You have successfully [petitioned](#) for an exception.

Non-Computer Science Core



In addition to the core CSE courses, BS CSE majors are expected to take the following supplemental courses:

Non-Computer Science Core

Course	Title	Credits	Prerequisites and Notes
ECE 2360*	Electronics for CSE majors	3	Must have credit for ENGR 1182.01, 1182.02, 1182.03, 1282.01H, 1282.02H, 1282.03H, or 1282.04H, or 1186 and 1187 and 1188 concurrent; and Math 1152, 1161.01, 1161.02, 1172, or 1181H; and Physics 1250, 1250H, or 1260, and CPHR 2.00 or above; and enrollment in CSE major. Not open to students with credit for 2020 or 2300.
ECE 2060	Introduction to Digital Logic	3	Not open to students with credit for ECE 2000, 2000.02, 2000.07, 2001, 2010, or 2017
Math 2568	Linear Algebra	3	Must have credit for 1172, 1544, 2153, 2162.xx, 2182H, or 4182H with a C- or higher, or have completed MATH 1152 and CSE 2321 with a C- or above in both.
Math 3345	Foundations of Higher Mathematics	3	Must have credit for CSE 2321 with a C- or higher; and have credit for 1152, 1161.xx, 1172, 1181H, 1534, 1544, or 4181H with a C- or higher.
Stat 3470	Introduction to Probability and Statistics for Engineers	3	Must have credit for Math 1152, 1161.xx, 1172, 1181H, or equiv, or permission of instructor. Not open to students with credit for 3440, 3450, 3450.01, 3450.02, or 3460.

*Required for students who entered the program in or after AU22. Students who entered prior to AU22 may take ECE 2020, but ECE 2360 is preferred.

Non-Computer Science Electives



In addition to the core outside department courses, BS CSE students must take 8 credits of math and/or science electives from the table below.

Math and Science Elective Options

Course	Title	Credits	Prerequisites and Notes
Math 2153	Calculus III	4	A grade of C- or above in 1152, 1172, 1534, 1544, 1181H, or 4181H, or credit for 153.xx, 154, 162.xx, or 162.01H. Not open to students with credit for any course 2153 or above, or for any quarter-system class 254.xx or above.
Math 2255 or Math 2415	Differential Equations and Their Applications or Ordinary and Partial Differential Equations	3	For Math 2255, must have a grade of C- or above in 2153, 2162.xx, 2173, 2182H, or 4182H, or credit for 254.xx, 263.xx, 263.01H, or 264H. Not open to students with credit for 2415, 5520H, 255, 415.xx, or 521H. For Math 2415, must have credit for Math 2153, 2162.xx, 2173, 2182H, 4182H, 254.xx, 263.xx, 263.01H, or 264H; or 2568 and either 1172 or 1544 (154). Not open to students with credit for 2255 (255), 5520H (521H), 2174, or 415.xx.
Stat 4201	Introduction to Mathematical Statistics I	4	
Stat 5301	Intermediate Data Analysis I	4	Must have credit for Math 1075 or equiv, or Math Placement Level of R, or permission of instructor. Not open to students with credit for 5302.
Anthrop 2200	Introduction to Physical Anthropology	4	
Biology 1113	Biological Sciences: Energy Transfer and Development	4	Must have credit for Math 1120, 1130, 1148, 1150, or above, or Math Placement Level L or M. Must have credit for or be taking Chem 1110, 1208, 1210, 1610, or 1910H, or permission of course coordinator. Not open to students with credit for 1113 or 1113.02.
Biology 1114	Biological Sciences: Form, Function, Diversity, and Ecology	4	Must have credit for Math 1120, 1130, 1148, or 1150 or above, or Math Placement Level L or M. Must have

Course	Title	Credits	Prerequisites and Notes
			credit for or be taking Chem 1110, 1208, 1210, 1610, or 1910H, or permission of course coordinator. Not open to students with credit for 1114 or 1114.02.
Chem 1210	General Chemistry I	5	Must have completed one unit of high school chemistry, and Math Placement Level L or M; or a grade of C- or above in Math 1120, 1130, 1131, 1148, 1150, or above. Not open to students with credit for 1220, 1620, 1920H, 1250, 1610, or 1910H.
Chem 1250	General Chemistry for Engineers	4	
EarthSc 1121	The Dynamic Earth	4	Must have credit for Math 1075 or above, or ACT Mathematics Subscore of 22 or higher, or Math Placement Level R or better. Not open to students with credit for 1100.
EarthSc 1122	Earth Through Time	4	
ENR 2100	Introduction to Environmental Science	3	
ENR 3000 and ENR 3001	Soil Science and Soil Science Laboratory	4	
FdScTe 2200	The Science of Food	3	Must have credit for Biology 1101 (101) or 1113 (113). Not open to students with credit for 201.
HCS 2201	Ecology of Managed Plant Systems	4	
HCS 2202	Form and Function in Cultivated Plants	4	Not open to students with credit for 300, 300E, 310, 315, or PlntBio 300.
Physics 1251	E&M, Optics, Modern Physics	5	Must have credit for Physics 1250, 1250H, 1260, or 1270; and Math 1141, 1151, 1154, or above; or permission of instructor. Must have credit for or be taking Math 1152, 1155, 1161, 1172, 1181H, or 4181H.

General Education



All students need 24 credits in general education. For details of how these hours are broken down, please refer to the [General Education](#) page for engineers. OSU's [class search engine](#) also features gen ed filters. For instructions on how to use these filters, refer to [this instructional video](#).

>> Additional Resources

Sample Schedule



Autumn	Spring
Year 1	
Engr 1100 - 1 credit	CSE 2221 (need C- or better) - 4 credits
Engr 1181 - 2 credits	Engr 1182 - 2 credits
Math 1151 - 5 credits	Math 1172 - 5 credits
Physics 1250 - 5 credits	GE-Writing - 3 credits
CSE 1223 - 3 credits	GE Launch Seminar - 1 credit
Total of 16 credits	Total of 15 credits
Year 2	
CSE 2231 - 4 credits	CSE 2331 - 3 credits
CSE 2321 - 3 credits	CSE 2421 - 4 credits
Stat 3470 - 3 credits	ECE 2060 - 3 credits
Math/Science Elective - 4 credits	Math 3345 - 3 credits
GE-Social/Behavioral - 3 credits	GE-Diversity - 3 credits
Total of 17 credits	Total of 16 credits
Year 3	
CSE 2431 - 3 credits	CSE 32X1 - 3 credits
CSE 390X - 4 credits	CSE 34X1 - 3 credits
ECE 2360 - 3 credits	CSE 35X1 - 3 credits
Math 2568 - 3 credits	CSE 2501 or PHILOS 2338 - 1 or 4 credits

Autumn	Spring
GE-History - 3 credits	[GE-Theme if not PHILOS 2338] - 4 credits
	Math/Science Elective - 4 credits
Total of 16 credits	Total of 17-18 credits
Year 4	
CSE 3341 - 3 credits	CSE 591X - 4 credits
Technical Elective - 3 credits	Technical Elective - 3 credits
Technical Elective - 3 credits	Technical Elective - 3 credits
Technical Elective - 3 credits	Technical Elective - 2 credits
GE-Lit/VPA - 3 credits	GE-Theme - 4 credits
Total of 15 credits	Total of 16 credits

Curriculum Sheet with Sample Schedule (download)

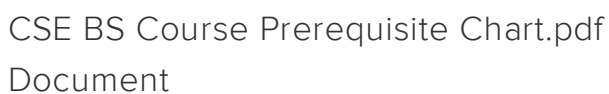


Curriculum Sheet with Sample Schedule (.docx)

Prerequisite Visualization



Capstone Pathway



A PDF printable version of the PNG image above.

Additionally, there is a **text-base description of this flowchart** designed with screen readers in mind.

A **graduation application** must be completed no later than the second Friday of the term before you wish to graduate. For example, students graduating in

Spring Semester will be required to submit their applications by the second Friday of the immediately preceding Autumn Semester.

Please note that a change in your graduating semester requires completion of a new application.

POCAT (Program Outcomes Achievement Test)

The Department of Computer Science and Engineering uses a comprehensive set of processes and tools to assess its programs to evaluate various aspects of the programs for the purposes of identifying and implementing improvements. To assist with this, during the graduation term, BS CSE, BS CIS, and BA CIS majors are required to complete the **exit test, POCAT**. Students will receive emails during their final semester about accessing and submitting the POCAT.

Standards of Academic Performance (Special Action Probation)



Special Action Probation is a departmental policy allowed by the university that assists with the review of grades at the department level. All CSE majors are subject to this policy for each term of enrollment.

Students may be placed on Warning or Special Action Probation depending on their academic situation: Warning or Special Action Probation for Grades, Warning or Special Action Probation due to Lack of Progress (LOP), or Special Action Probation after Reinstatement.

Placement on Warning

Students placed on Warning will receive a notice, but no action will be placed on the transcript.

Warning for Grades:

A Warning will be issued for the first term in which either of the following conditions apply:

- failure of one or more major or required non-major courses,
- a mark of Incomplete (I) with a lapse grade of E for one or more major or required non-major courses.

Warning for LOP

A Warning will also be issued if a pattern of W or I marks is apparent:

- two consecutive semesters of enrollment of W or I marks in one or more major or required non-major courses.

Removal of Warning

Students will be required to consult with their assigned academic advisor to discuss options that will lead to academic success, take at least one major or required non-major course, and must not earn marks of E, W, or I in any major or required non-major courses in the following term of enrollment.

Placement on Special Action Probation

Students may be placed on Special Action Probation if they do not satisfy the conditions for removal of a warning or if they meet other conditions that lead directly to Special Action Probation.

- **SAP due to Grades:**

- If a student on warning subsequently fails one or more major or required non-major courses, which may or may not be the same course that was failed previously, the student will be placed on formal Special Action Probation, which will be noted on the transcript.
- For students on warning due to Incompletes ("I") with an alternate grade of "E": If an "I" converts to "E" and the student subsequently earns one or more failing grades in major or required non-major courses, the student will be placed on formal Special Action Probation.

- **SAP due to LOP:** SAP due to LOP applies to students who are not making adequate or appropriate progress toward graduation in one or more of the following manners:

- Three or more consecutive semesters of enrollment of "W" or "I" marks in any degree-applicable courses that prevent the student from making academic progress toward graduation will result in being placed on formal Special Action Probation.
- Three or more unsuccessful attempts in the same major or required non-major course will result in being placed on formal Special Action Probation.
- No degree-applicable courses for two consecutive terms of enrollment will result in being placed on formal Special Action Probation.

- **SAP after Reinstatement:** Students reinstated to the CSE major are automatically reinstated on Special Action Probation.

Removal from Special Action Probation:

- **SAP due to Grades and SAP after Reinstatement:**
 - Students must consult with their assigned academic advisor to determine coursework that will lead to academic success.
 - Students must take at least one major or required non-major course in the following term of enrollment and earn a passing grade.
- **SAP due to LOP:**
 - Students must earn passing grades in degree-applicable courses in the next term of enrollment.
 - Students must consult with their assigned academic advisor to discuss the circumstances that led to their situation and to determine appropriate next steps.

Conditions for Department Dismissal and College Dismissal

Students who do not meet the conditions to be removed from Special Action Probation are eligible to be dismissed from the department. Students whose CGPA is below 2.00 may also be eligible for dismissal from the College of Engineering.

Conditions for Reinstatement

Students who have been dismissed from the department or college and have been away for at least two semesters may seek reinstatement to the CSE major by filing a Petition for Reinstatement. Consideration of reinstatement earlier than two semesters after dismissal may be done on a case-by-case basis. Students interested in reinstatement are expected to consult with an academic advisor prior to submission of the petition.

Questions regarding Warning and Special Action Probation may be emailed to the Computer Science Undergraduate Advising office at compsciadvising@osu.edu.

> Department and Program Accreditations

ABET



The BS CSE program is accredited by the Engineering Accreditation Commission of **ABET**, under the General Criteria and the Program Criteria for Electrical, Computer, Communications, Telecommunication(s) and Similarly Named Engineering Programs.

The BS CSE program is also accredited the Computing Accreditation Commission of **ABET**, under the General Criteria and the Program Criteria for Computer Science and Similarly Named Computing Programs.

NSA Cybersecurity



The CSE department has been designated by the **NSA as a Center for Academic Excellence in Cybersecurity (CAE-C)**.

The BS CSE program with a Specialization in Information and Computation Assurance is recognized as an **NSA Validated Cyber Defense (NSA CAE-CD)** Program of Study.

Additional Information



For additional details and information of the College of Engineering's various accreditations, please visit the [College of Engineering's accreditations page](#).



THE OHIO STATE UNIVERSITY
COLLEGE OF ENGINEERING

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Computer Science and Engineering

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*Ohio State
is in the
process of
revising
websites
and program
materials to
accurately
reflect
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with the law. While this work occurs, language referencing protected class status or other activities prohibited by Ohio Senate Bill 1 may still appear in some places. However, all programs and activities are being administered in compliance with federal and state law.